

Sri Lankan Black Pepper

Crop Establishment

Knowledge Domain: 01 — Planting Practices (Crop Establishment)

Project: RAG Chat System for Sri Lankan Black Pepper Knowledge

This document covers the full crop establishment process for black pepper (*Piper nigrum*) cultivation in Sri Lanka, from the selection and preparation of planting material through nursery management and field planting.

1. Planting Material

Black pepper is propagated vegetatively using stem cuttings rather than by seed, ensuring genetic consistency and predictable yield performance. The quality of planting material is the foundation of a productive plantation.

1.1 Propagation Method

- **Standard Method:** Black pepper is almost exclusively propagated vegetatively from stem cuttings — seed propagation is not used for commercial cultivation.
- **Cutting Source:** For commercial cultivation, cuttings are taken from terminal stems (top growing tip of the vine) or from ground runners (rooted runners along the soil surface).
- **Lateral Cuttings:** Cuttings taken from lateral branches will produce bush-type plants rather than climbing vines — this is generally not preferred for commercial plantations.

1.2 Mother Vine Selection

The quality of cuttings is entirely dependent on the quality of the mother vine. Strict selection criteria must be applied:

Selection Criterion	Standard / Requirement
Yield	High-yielding vine with consistent production over multiple seasons
Health Status	Completely free from pest infestation and disease symptoms
Vine Vigour	Vigorous, healthy growth with strong climbing habit
Lateral Branches	Produces abundant lateral branches with short inter-nodal distances
Spike Quality	Long spikes with complete coverage by berries (no gaps)
Berry Quality	Bold, well-filled berries with good dry weight ratio
Climate Tolerance	Tolerant to the specific climatic conditions of the intended planting area

Key Rule: Never take cuttings from a vine showing any signs of disease, pest damage, or poor vigour — these defects will carry into the new plants.

2. Nursery Management

A structured nursery phase of 4 to 6 months is essential for producing field-ready plants. Rushing the nursery process is one of the most common causes of poor field establishment.

2.1 Nursery Media and Containers

- **Poly Bags:** Use 250-gauge polythene bags measuring 8 inches × 5 inches.
- **Growing Medium:** Fill bags with a mixture of equal parts topsoil, cow dung, coarse sand, and coir dust. This blend provides good drainage, moisture retention, and organic nutrition.
- **Mix Quality:** Ensure the mixture is well-mixed and free from soil-borne pathogens before use.

2.2 Nursery Timeline

The nursery process follows a structured 6-phase timeline from planting to field readiness:

Nursery Phase	Timing	Action / Notes
Planting in Poly Bags	Week 1	Insert cuttings into 250-gauge 8"×5" poly bags filled with topsoil, cow dung, sand, and coir dust (equal parts). Place in propagator.
Propagator Period	Weeks 1–4	Keep sealed in propagator to maintain high humidity for root development. No direct watering needed.
Partial Opening	Weeks 4–7	Partially open propagator for controlled watering and weeding. Begin acclimatisation.
Propagator Removal	Weeks 6–7	Gradually remove polythene cover over 2–3 weeks. Do not remove abruptly — stagger removal to avoid transplant shock.
Shade House Hardening	Weeks 7–12	Move plants to shade house. Practice progressive hardening by reducing shade over the final 3–4 weeks.
Field-Ready Plants	Month 4–6	Plants should have 5–8 healthy leaves and vigorous growth before field planting. Total nursery period: 4–6 months.

Critical Point: The total nursery period must be 4 to 6 months. Removing plants too early (before they have 5–8 leaves and strong roots) results in high field mortality. Hardening in the final 3–4 weeks is essential before transplanting.

2.3 Hardening Process

Hardening is the gradual acclimatisation of nursery plants to outdoor conditions before field planting:

1. After removing the propagator, place plants in a well-ventilated shade house.
2. Over 3–4 weeks, progressively reduce the shade level — start at 75% shade and reduce to 25% by the final week.

3. Plants are considered field-ready when they show vigorous growth, 5–8 fully developed leaves, and a firm root ball.

3. Support Systems for Pepper Vines

Pepper vines are climbing plants that require a support structure to grow vertically. In Sri Lanka, live support trees are the standard practice, providing shade, structural support, and additional organic matter through leaf litter.

3.1 Live Support Trees

Three species are used as live standards for pepper vines in Sri Lanka:

Species	Common Name	Usage	Notes
Gliricidia sepium	Gliricidia	Primary / Most Common	Fast-growing; nitrogen-fixing legume. Lopped leaves used as green mulch. Sticks: 3–5 cm diameter, 2.5 m length, planted 20 cm deep.
Erythrina indica	Dadap	Secondary / Alternative	Provides good shade and structural support. Commonly used in older plantations.
Grevillea robusta	Silky Oak	Alternative	Tall timber tree that provides durable long-term support and partial shade.

3.2 Gliricidia Support — Installation Guidelines

- **Stick Dimensions:** Select Gliricidia sticks of 3–5 cm diameter and 2.5 m in length.
- **Planting Depth:** Plant the Gliricidia stick at a depth of 20 cm at the corner of the planting pit, firmly embedded to withstand vine weight.
- **Lead Time — Critical:** Gliricidia support sticks must be planted at least 6 months before the pepper vine is planted. This allows the support to establish roots and provide adequate shade before pepper planting.
- **Ongoing Management:** Lop the Gliricidia canopy twice per year to prevent overshading. Lopped leaves and branches make excellent green mulch at the base of the vine.

Why Gliricidia? *Gliricidia sepium* is a nitrogen-fixing legume, meaning it improves soil fertility naturally. Its fast growth, easy propagation from cuttings, and tolerance to lopping make it ideal as a long-term pepper support in Sri Lanka's agro-climatic conditions.

4. Field Planting

Field planting must be timed, spaced, and executed correctly to ensure strong vine establishment. Poor planting technique at this stage is difficult to correct later.

4.1 Field Planting Quick Reference

All critical field planting specifications are summarised below:

Parameter	Specification / Guideline
Recommended Spacing	2.4 m × 2.4 m (mono-crop and intercrop with coconut)
Planting Density	1,700 plants per hectare
Pit Dimensions	60 cm × 60 cm × 60 cm
Pit Fill Mix	Topsoil + cow dung or compost (well mixed)
Support Stick Dimensions	3–5 cm diameter, 2.5 m length (Gliricidia)
Support Planting Depth	20 cm into the ground at the corner of the planting pit
Support Lead Time	At least 6 months before pepper planting
Plant Age at Field Planting	4–6 months old nursery plants
Plant Condition	5–8 healthy leaves, vigorous growth
Distance from Support	15–20 cm away from the support stick
Planting Season	Onset of monsoon rains
Post-Planting Actions	Provide temporary shade; apply mulch to base to conserve soil moisture

4.2 Land Preparation and Pit Filling

- **Land Preparation:** Prepare the field by clearing weeds, breaking up compacted soil, and improving drainage where necessary.
- **Pit Dimensions:** Dig planting pits of 60 cm × 60 cm × 60 cm at the marked spacing of 2.4 m × 2.4 m.
- **Pit Fill:** Fill pits with a mixture of topsoil and well-composted cow dung or compost to a level that allows the plant to sit at the correct depth.
- **Settling Time:** Allow filled pits to settle for at least 2 weeks before planting.

4.3 Planting Procedure

4. Plant at the onset of monsoon rains — this is the single most critical timing factor for successful establishment.
5. Select only 4–6 month old, hardened nursery plants with 5–8 healthy leaves and vigorous growth for field planting.
6. Plant the pepper vine 15–20 cm away from the base of the Gliricidia support stick — do not plant directly at the base.
7. Firm the soil around the root ball gently to eliminate air pockets without compacting the soil.
8. Immediately after planting, erect temporary shade using coconut fronds, shade cloth, or any suitable material to protect the young plant from direct sunlight.
9. Apply a layer of mulch (straw, coir, or Gliricidia cuttings) around the base of the vine in a 50 cm radius to conserve soil moisture and suppress weeds.

Intercrop Note: *The recommended spacing of 2.4 m × 2.4 m is suitable for both mono-crop plantations and intercropping with coconut. This spacing yields approximately 1,700 plants per hectare and allows sufficient light and air circulation for both crops.*

Planting Season Warning: *Never plant pepper vines during dry periods. Without reliable soil moisture from monsoon rains, young transplants suffer severe water stress and high mortality. If planting outside the monsoon window is unavoidable, supplemental irrigation must be in place before planting.*